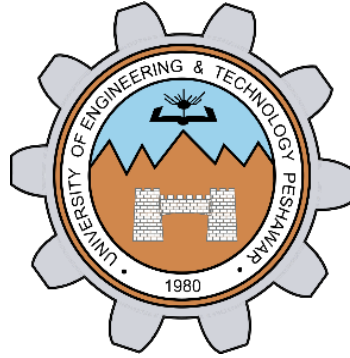




**UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR
(NOWSHERA CAMPUS)**

**COMPUTER
PROGRAMMING**

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Lab Report

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Program No: 1

```
#include <iostream>
using namespace std; int
main() { int arr[10], max,
min, sum = 0;

// Taking input from user
cout << "Enter 10 integers:\n";
for(int i = 0; i < 10; i++) { cin >>
arr[i]; }

// Initialize max and min with first element max
= min = arr[0];

// Loop to find max, min and sum
for(int i = 0; i < 10; i++) { if(arr[i] > max) max
= arr[i]; // Update max if(arr[i] < min) min =
arr[i]; // Update min sum += arr[i];
// Add to sum
}

// Display results cout << "Maximum: " <<
max << endl; cout << "Minimum: " << min
<< endl; cout << "Average: " << sum /
10.0 << endl;

return 0;
}
```

Program No: 2

```
#include <iostream>
using namespace std;

// Function to reverse array void reverseArray(int
arr[], int size) { for(int i = 0; i
< size/2; i++) { swap(arr[i], arr[size - i - 1]); // Swap
elements
}
}
```

```

int main() { int arr[5] =
{1,2,3,4,5}; // Display
original array cout <<
"Before Reversal: ";
for(int i=0;i<5;i++)
cout << arr[i] << " ";

reverseArray(arr, 5); // Function call

// Display reversed array cout <<
"\nAfter Reversal: "; for(int i=0;i<5;i++)
cout << arr[i] << " ";

return 0;
}

```

Program No: 3

```

#include <iostream>
using namespace std;

```

```

int main() { int A[3][3],
B[3][3], C[3][3];

```

```

// Input Matrix A cout <<
"Enter Matrix A:\n"; for(int
i=0;i<3;i++) for(int j=0;j<3;j++)
cin >> A[i][j];

```

```

// Input Matrix B cout <<
"Enter Matrix B:\n"; for(int
i=0;i<3;i++) for(int j=0;j<3;j++)
cin >> B[i][j];

```

```

// Add matrices for(int
i=0;i<3;i++) for(int
j=0;j<3;j++)
C[i][j] = A[i][j] + B[i][j];

```

```

// Display result cout <<
"Result Matrix:\n"; for(int
i=0;i<3;i++) { for(int

```

```
j=0;j<3;j++) cout << C[i][j]
<< " "; cout << endl;
}
```

```
return 0;
}
```

Program No: 4

```
#include <iostream>
using namespace std;
```

```
int main() { string
names[5];
```

```
// Input names cout << "Enter
5 names:\n"; for(int
i=0;i<5;i++) cin >> names[i];
```

```
// Sorting using simple comparison for(int
i=0;i<5;i++) { for(int j=i+1;j<5;j++) {
if(names[i] > names[j]) { swap(names[i],
names[j]); // Swap names
}
}
}
```

```
// Display sorted names cout
<< "Sorted Names:\n"; for(int
i=0;i<5;i++) cout << names[i]
<< endl;
```

```
return 0;
}
```

Program No: 5

```
#include <iostream>
using namespace std; //
```

Function to calculate

```
total load double
totalLoad(double arr[],
int size) { double sum =
```

```

0; for(int i=0;i<size;i++)
sum += arr[i]; return
sum;
}

```

```

// Function to find peak load double
peakLoad(double arr[], int size) { double
max = arr[0]; for(int i=1;i<size;i++)
if(arr[i] > max) max = arr[i]; return max;
}

```

```

int main() { double
load[24];

```

```

// Input load data cout << "Enter 24-
hour load:\n"; for(int
i=0;i<24;i++) cin >> load[i];

```

```

// Display results cout << "Total Load: " << totalLoad(load,24)
<< endl; cout << "Peak Load: " << peakLoad(load,24) << endl;

```

```

return 0;
}

```

Program No: 6

```

#include <iostream>
using namespace std;

```

```

// Function to find topper void topper(string
names[], int marks[], int size) { int max = marks[0],
index = 0; for(int i=1;i<size;i++) { if(marks[i] > max)
{ max = marks[i]; index = i;
}
}
cout << "Topper: " << names[index] << endl;
}

```

```

// Function to display pass/fail void passFail(string
names[], int marks[], int size) { for(int
i=0;i<size;i++) { cout << names[i] << " - "; if(marks[i]
>= 50) cout << "Pass\n"; else cout <<
"Fail\n";
}
}

```

```
}  
}
```

```
// Function to search student void search(string  
names[], int size, string key) { for(int  
i=0;i<size;i++) { if(names[i] == key) { cout  
<< "Found: " << names[i] << endl; return;  
}  
}  
cout << "Not Found\n";  
}
```

```
int main() { string names[3] =  
{"Ali", "Ahmed", "Sara"}; int marks[3] =  
{80,45,90};
```

```
topper(names, marks, 3);  
passFail(names, marks, 3);  
search(names, 3, "Ahmed");
```

```
return 0;  
}
```

Program No: 7

```
#include <iostream>  
using namespace std; //  
Calculate total energy  
double  
totalEnergy(double  
arr[]) { double sum=0;  
for(int i=0;i<24;i++)  
sum+=arr[i]; return sum;  
}
```

```
// Find hour of maximum generation  
int maxHour(double arr[]) { int  
index=0; for(int i=1;i<24;i++) if(arr[i] >  
arr[index]) index = i; return index;  
}
```

```

int main() { double
energy[24];

//    Input data
for(int i=0;i<24;i++) cin
>> energy[i];

    cout << "Total Energy: " << totalEnergy(energy) << endl;
    cout << "Max Hour: " << maxHour(energy) << endl; cout
<< "Average: " << totalEnergy(energy)/24 << endl;

return 0;
}

```

Program No: 8

```

#include <iostream>
using namespace std;

```

```

int    main() { double
v[20]; int low=0, high=0;

```

```

// Input voltage readings for(int
i=0;i<20;i++) cin >> v[i]; // Check safe
limits for(int i=0;i<20;i++) { if(v[i] <
3.0) low++; if(v[i] > 4.2) high++;
}

```

```

// Find max and min voltage double
max=v[0], min=v[0];
for(int i=1;i<20;i++) { if(v[i] >
max) max=v[i]; if(v[i] < min) min=v[i];
}

```

```

cout << "Low: " << low << endl; cout <<
"High: " << high << endl; cout <<
"Max: " << max << endl; cout <<
"Min: " << min << endl;

```

```

// Warning if unsafe if(low>0 || high>0)
cout << "Warning: Unsafe readings!\n";

```

```
return 0;
}
```

Program No: 9

```
#include <iostream>
using namespace std;
```

```
int main() { int arr[4][7];
```

```
// Input data for(int
i=0;i<4;i++) for(int
j=0;j<7;j++) cin >>
arr[i][j];
```

```
// Calculate total per region
for(int i=0;i<4;i++) { int sum=0;
for(int j=0;j<7;j++)
sum+=arr[i][j]; cout <<
"Region " << i+1 << " Total: "
<< sum << endl;
}
return 0;
}
```

Program No: 10

```
#include <iostream>
using namespace std;
int main() { int
att[5][7];
```

```
// Input attendance
for(int i=0;i<5;i++)
for(int j=0;j<7;j++) cin
>> att[i][j];
```

```
int max=0, index=0;
```

```
// Calculate attendance
for(int i=0;i<5;i++) { int
sum=0; for(int
```



```
j=0;j<7;j++)  
sum+=att[i][j];
```

```
cout << "Student " << i+1 << ": " << sum << endl;
```

```
if(sum > max) { max  
= sum; index  
= i;  
}  
}
```

```
cout << "Highest Attendance: Student " << index+1 << endl; return  
0; }
```